

LAGOS UNIVERSITY TEACHING HOSPITAL ENTRANCE EXAMINATIONS 2022

TIME:45MIN

- Without words he accused him directly of treachery.
A.amending B.modifying C.mixing D.mincing E.minimizing
- They tried to cash in the people ignorance A.under B.on
C.against D.with E.at
- Always remember to.....the lights before leaving the room
A.put on B.put off C.switch off D.blow out E.remove
- I am disappointed the ways you conducted yourself at the party.A.by B.for C.due D.in E.at
- The proprietors should be blamed for such a deplorable condition in the nursery schools.....A.isn't it B. shouldn't they C.should they D.isit E.wouldn't they
- Do you mind.....another minute or two?A.to wait B.wait C.waiting D.being waited F.waits
- It had been raining before the match started.....A.isn'tit B.hasn't it C.hadn't it D.wasn't it E.haven't it
- Mary goes to school.....bus A.in B.on C.with D.of E.by
- The plane overshot the.....in a minor accident A.read B.hanger C.runway D.tarmac F.railway
- Journalists always collect and publish.....A.information B.informations C.an information D.much information E. every information
- Amicrometer is defined as one millioneth of a millimeter. Alength of 12,000 micrometer may be represented as?A. 0.00012M B. 0.0000012M C. 0.000012M D. 0.00000012M F. 0.000000012M
- Factorize $3x^3+4x^2-13x+6$ completely.given that x-is a factor. A.(x-1)(x-3)(x+2)B.(x-1)(x+3}{(x-2)C(x1)(x+3)(3x-2)D.(x-1)(x+3)(3x-2)2.(x-IXx-3}{(x+2)
- If the price of oranges was raised by %k per orange,the number of oranges a customer can buy for N2.40 will be less by 16.what is the price of an orange?A.21/2k B.31/2k C.51/2k D. 20k 2. 25k
- Find all real numbers x which satisfies the inequality $\frac{1}{3}(x+1) > \frac{1}{5}(x+4)$.A.x<11B.x<-1C.x>6 D.11E.x>-6
- 7 pupils df average age 12 years leave a class of 25 pupils average age 14years.If 6 new pupils of average age 11 years join the dass, what is the average age of the pupils now in the class?A.13 years B.12 years 11/2 months C.13 years 5 months D.13 years 10 months E.13 years 71/2 months
- Given a regular hexagon,calculate each interior angle of the hexagon A.60° B.120C. 45° 0. 135° 2. 140°
- Without using tablessolve the equaticn: $Bx^2=2/5$.A.4B.6C.8 D. 10E. 12
- A student measures a piece of rope and found that it was 1 .26m long.If the actual length of the rope is1.25m,what was the percentage error in the measurement?A.0.40%B.0.01% C. 0. 25%0. 0. 89%2. 0. 80%
- Express the product of 0.21 and 0.34 in standard form A.7.14 $\times 10^3$ B. 7. 14x10¹ c. 7. 14x10² D. 7. 14 $\times 10^4$ E7. 14 $\times 10^5$
- If 5,8,6 and 2 ocur with frequendes 3,2,4 and 1 respectively. find the product of the modal and median number A.36 B.48 C.300.402.None of the above
- A plant which grows on another plant without apparent harm to the host plant is called A.a parasite B.an epiphyte C.a saprophyte D.a predator E.a hermaphrodite
- Which of these types af skeleton is most appropriate to the cockroach?A.hydrostatic skeleton B.exoskeleton C. endoskeleton D.cartilaginous skeleton E.bony skeleton
- Which of these is not true of the insed?The possession of A. two pairs of antennae B.jointed appendages C.exoskeleton D. three pairs of legs E.segmented body
- All living things A.photosynthesis B.respire C.move D.transpire E.feed
- A secchi disc is used in the detemination of A.rainfall B.tides C.waves D.turbidity E.current velocity
- Which of the following is not an excretory product?A.urine B.sweat C.feaces D.salts E.carbon dioxide
- Which of the following organs produce bile?A gallbladder B.pancreas C.spleen D.liver E.stomach
- Which of the following food substances is digested in the stomach?A.carbohydrates B.fats and oil C.fats and proteins D.proteins E.carbohydrates and fats
- For polination and fruit formaton,the esential part(s) of the flower should be the A.corolla B.ovary C.pistil (gynoecium's) D.ovules E.receptade
- Which part of the human brain is concerned with reflexes controlling the rate of heartbeat and breathing?A.medulla B.cerebrum C.cerebellum 0.pineal body E.olfactory lobe
- Which of the follbwing is not a conequene of a force filed?A. weight B.surface tension C.gravitational pull D.magnetic force E.electric force
- Which of the following is used to determine therelhtive density of the acid in a car battery?A.hypsometer B. hygrometer C.manometer D.hydrometer E.nanometer
- The motion of the prongs of sounding tuning fork is?A random B.translational C.rotational D.vibration E vibratory and rotational
- A simple microscapse forms an image twice the size of an object If the focal length of the lens of microscope is 29cm,how far is the object from the lens?A.10m B.20m C.30m 0.40m E.60m
- An avocado fruit drops from the top of a tree 45m tall.How long does it take to reach the ground?A.3.Os B.4.5s C.6.Os D.8.6s E.9.Os
- Which of the following is a scalar quantity?A.momertum B.Acceleration C.displacement D.Distance E.force
- A ball bearing is gently released from rest and allowed to fall through a viscous fluid.Which of the following statements about the motion is correct?A.its acceleration decreases before terminal velocity is attained B.when terminal velocity is attained the acceleration of the fluid becomes zero C.its velocity increases before terminal velocity is attained D.there is no resultant force on the ball before it attains terminal velocity

38. When the vapour of a substance is in equilibrium with its own liquid, it is said to be A. gaseous B. unsaturated C. saturated D. diffused E. liquefied
39. A man standing between two parallel mirrors in a barbers shop will see the following number of his own image A. eight B. two C. four D. one 2. Infinite
40. If the wave length of a wave traveling with a velocity of 360 m/s is 60 cm, the period of the wave is A. 6 s B. 3.5 s C. 0.17 s D. 0.61 s E. 3 s
41. 120 cm³ of hydrogen were sparked with 60 cm³ of oxygen at 110°C. What was the volume of steam produced? The equation for the reaction is: $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{g})$ A. 30 cm B. 60 cm C. 90 cm D. 120 cm E. 150 cm
42. Catalytic hydrogenation of oils results in the production of A. soaps B. detergents C. alkanes D. margarine E. butter
43. Which of the following compounds will undergo additional reaction? A. ethyne B. butane C. pentane D. ethanol E. tetrachloromethane
44. The products of the electrolysis of dilute sodium chloride solution with platinum electrodes are A. hydrogen and oxygen B. oxygen and chlorine C. chlorine and water D. sodium amalgam and chlorine E. sodium hydroxide and water
45. When starch undergoes complete enzyme-catalysed hydrolysis, the resulting product is A. glucose B. maltose C. sucrose D. fructose E. cellulose
46. Compounds that have the same molecular formula but different structures are said to be A. allotropic B. polymorphic C. polymeric D. isomeric E. isotropic
47. The maximum number of electrons that can be accommodated in the shell having the principal quantum number 3 is A. 3 B. 9 C. 10 D. 18 E. 32
48. The following acids are non-basic except A. methanoic acid B. dioxynotrate C. ethanedioic acid D. oxochlorate I add E. hydrobromic acid
49. What is the quantity of electricity produced when a current of 0.5 A is passed for 5 hours 45 mins? (F = 96500 C) A. 0.11 F B. 0.12 F C. 0.22 F D. 1.1 F E. 2.2 F
50. Which of the following pH values is likely to be that of a close slightly alkaline system? A. 2 B. 5 C. 8 D. 11 E. 13

SOLUTION TO LUTH SCHOOL OF NURSING 2022 EXAM

1. A	2. C	3. C	4. E	5. B	6. C	7. C	8. E	9. C	10. A
11. C	12. D	13. B	14. D	15. D	16. B	17. A	18. E	19. C	20. E
21. B	22. B	23. A	24. E	25. D	26. C	27. D	28. C	29. C	30. A
31. B	32. D	33. D	34. **	35. A	36. D	37. C	38. C	39. E	40. N/A
41. D	42. D	43. A	44. A	45. A	46. D	47. D	48. C	49. C	50. D

LUTH SCHOOL OF NURSING PAST QUESTIONS AND ANSWERS

TEST 2021

TIME: 45 MIN

- If Feyinwa found that thieves had entered her house in her absence. She went to the police to report the A. breakout B. breakup C. break in D. breakthrough E. breakin
- Gone are the days when he Enjoy patronage A. would B. will C. could D. used to E. can
- All Well with Peter. A. are not B. have not been C. were not D. is not E. aren't
- The villagers looked their leader for good examples. A. upto B. onto C. up at D. forward to E. into
- Emeka failed because the examination was difficult for him. A. so B. very C. highly D. too E. much
- The dancers were all in before their departure. A. good spirits B. good spirit C. high spirit D. high spirits E. high spirit
- The suspect defrauded his victim of large sums of money. A. unsuspected B. unsuspecting C. unexpected D. unexpecting E. suspecting
- The judge acquitted the accused on all the eight counts. A. of B. in C. from D. upon E. on
- My uncle is one of the of society A. elitists B. elites C. elite D. elitist
- On all probability, the train will arrive today. A. in B. under C. for D. by E. upon
- Which of the following solids has a network structure? A. diamond B. iodine C. sulphur D. graphite E. butter
- The following gases decolorize bromine water except A. C₂H₂ B. C₂H₄ C. C₂H₆ D. C₃H₈ E. C₄H₁₀
- B. oxygen and chlorine C. chlorine and water D. sodium amalgam and chlorine E. sodium hydroxide and water
- When starch undergoes complete enzyme-catalysed hydrolysis, the resulting product is A. glucose B. maltose C. sucrose D. fructose E. cellulose
- Compounds that have the same molecular formula but different structures are said to be A. allotropic B. polymorphic C. polymeric D. isomeric E. isotropic
- The maximum number of electrons that can be accommodated in the shell having the principal quantum number 3 is A. 3 B. 9 C. 10 D. 18 E. 32
- The following acids are non-basic except A. methanoic acid B. dioxynotrate C. ethanedioic acid D. oxochlorate I add E. hydrobromic acid
- What is the quantity of electricity produced when a current of 0.5 A is passed for 5 hours 45 mins? (F = 96500 C) A. 0.11 F B. 0.12 F C. 0.22 F D. 1.1 F E. 2.2 F
- Which of the following pH values is likely to be that of a close slightly alkaline system? A. 2 B. 5 C. 8 D. 11 E. 13

24. A baker used 40% of a 50kg bag of flour. If 1/8 of the amount used was for cake, how many kilograms of flour was used for cake? A. 2% B. 64% C. 17% D. 19 1/3
25. Find the average of the first four prime numbers greater than 10. A. 20 B. 19 C. 17 D. 15 E. 13
26. Find the mean deviation of 6, 7, 8, 9, 10. A. 1.2 B. 1.5 C. 2 D. 8 E. 10
27. The variance of a given distribution is 25, what is the standard deviation? A. 125 B. 75 C. 5 D. 5 E. 3
28. Express 7/19 as a percentage, correct to 1 decimal place. A. 2.7% B. 3.7% C. 27.1% D. 36.8% E. 42.2%
29. Given that $\log_4 x = -3$, find x. A. 1/81 B. 1/640.64 D. 881 E. 102
30. A chord of a circle radius 26cm is 10cm from the center of the circle. Calculate the length of the chord. A. 16cm B. 27.86cm C. 32cm D. 40cm E. 48cm
31. Which of the following hormones is produced during fright or when agitated? A. Insulin B. adrenalin C. thyroxine D. pituitrin
32. Which of the following animals is cold-blooded? A. cat B. lizard C. whale D. bird E. dog
33. Spirogyra reproduces vegetatively by A. spore production B. fragmentation C. multiple fission D. binary fission E. division
34. All the following are digestive enzymes except A. bile B. lipase C. maltase D. pepsin E. ptyalin
35. If a 26-year-old man married a one-eyed woman and they had four children, how many of them would be blind like their father? A. all B. 3 C. 2 D. 1 E. none
36. A tapeworm has no alimentary canal because A. it is an autotrophic B. it does not feed C. it has no enzymes D. its body absorbs digested food E. it is long
37. Where is energy produced in the cell? A. nucleus B. nucleolus C. lysosomes D. mitochondria
38. Which of the following structures is not found in a female agama lizard? A. pre-anal pads B. eardrums C. nuchal crest D. gular fold E. chloroplast
39. Which of the following diseases is NOT caused by a virus? A. rinderpest B. maize rust C. Newcastle disease D. swine fever E. nasal scale
40. Plants which can survive in places where the water supply is limited are A. Bryophytes B. mesophytes C. xerophytes D. hygrophytes E. pteridophytes
41. Which of the following types of waves cannot travel through a vacuum? A. sound waves B. light waves C. infra-red waves D. x-radiation E. radio waves
42. The temperature at which the water vapour in the air saturates the air and begins to condense is known as A. boiling point B. melting point C. triple point D. dew point E. critical temperature
43. The motion of the moving skin of a talking drum can rightly be described as A. translational B. random C. rotational D. oscillatory E. transitory
44. What is the relative permittivity of a capacitor if its capacitance with a medium as dielectric is 16 farads, and its capacitance with vacuum as dielectric is 2 farads? A. -1 B. -% C. 2 D. 6 E. 8
45. The activity of a radioactive substance depends on A. temperature and purity B. temperature and age C. age, purity and temperature D. purity and age E. none of the above
46. The principle of the transmissibility of pressure in fluids at rest in all directions is known as A. Archimedes principle B. Floatation principle C. Newton's law D. Pascal's law E. Boyle's law
47. Change of state is accompanied by change of A. temperature B. volume C. heat content D. temperature and volume E. volume and heat content
48. The lack of power of accommodation which is mainly due to the hardening of the eye is called A. myopia B. hypermetropia C. presbyopia D. eye ring E. astigmatism
49. The hatch door of a submarine has an area of 0.5m². The specific gravity of sea water is 1.03. (assume that g = 10/s, and neglect the atmospheric pressure). The force exerted by sea water on the hatch door at a depth of 200m is A. 1.03 x 10⁵N B. 1.03 x 10³ N C. 1.06 x 10⁵N D. 2.06 x 10⁶N E. 1.03 x 10⁹ N/m²
50. The point beyond which a stretching spring does not return to its original length is called A. breaking point B. elastic limit C. spring constant D. elastic point E. release point

SOLUTION TO LUTH SCHOOL OF NURSING 2021 EXAM

1. C	2. D	3. D	4. A	5. D	6. B	7. C	8. A	9. B	10. B
11. A	12. A	13. A	14. A	15. D	16. C	17. E	18. A	19. B	20. A
21. C	22. *	23. D	24. A	25. D	26. D	27. D	28. E	29. B	30. E
31. B	32. C	33. B	34. A	35. A	36. D	37. D	38. E	39. B	40. C
41. A	42. D	43. D	44. C	45. C	46. D	47. C	48. C	49. B	50. B

- had four children, how many of them would be blind like their father? A. all B. 3 C. 2 D. 1 E. none
36. A tapeworm has no alimentary canal because A. it is an antitropic B. it does not feed C. it has no enzymes D. its body absorbs digested food E. it is long
37. Where is energy produced in the cell? A. nucleus B. nucleolus C. Lysosomes D. mitochondria
38. Which of the following structure is not found in a female agama lizard? A. pre-anal pads B. eardrums C. nuchal crest D. gular fold E. chloroplast
39. Which of the following disease is NOT caused by a virus? A. rinderpest B. maize rust C. Newcastle disease D. swine fever E. nasal scale

B. volume C. heat content D. temperature and volume E. volume and heat content

48. The lack of power of accommodation which is mainly due to the hardening of the eye is called A. myopia B. hypermetropia C. presbyopia D. eye ring E. astigmatism
49. The hatch door of a submarine has an area of 0.5m^2 . The specific gravity of sea water is 1.03. (assume that $g = 10/\text{s}$, and neglect the atmospheric pressure). The force exerted by sea water on the hatch door at a depth of 200m is A. $1.03 \times 10^5\text{N}$ B. $1.03 \times 10^3\text{N}$ C. $1.06 \times 10^5\text{N}$ D. $2.06 \times 10^6\text{N}$ E. $1.03 \times 10^0\text{N/m}^2$
50. The point beyond which a stretching spring does not return to its original length is called A. breaking point B. elastic limit C. spring constant D. elastic point E. release point

SOLUTION TO LUTH SCHOOL OF NURSING 2021 EXAM

1. C	2. D	3. D	4. A	5. D	6. B	7. C	8. A	9. B	10. B
11. A	12. A	13. A	14. A	15. D	16. C	17. E	18. A	19. B	20. A
21. C	22. *	23. D	24. A	25. D	26. D	27. D	28. E	29. B	30. E
31. B	32. C	33. B	34. A	35. A	36. D	37. D	38. E	39. B	40. C
41. A	42. D	43. D	44. C	45. C	46. D	47. C	48. C	49. B	50. B

LUTH SCHOOL OF NURSING PAST QUESTIONS AND ANSWERS

TEST 2020

TIME: 45 MIN

1. I have the of meeting him A. privilege B. privilege C. previledge D. priviledge E. privileges
2. My price for the shoes is fifty naira I cannot anything less than that. A. bear with B. settle for C. agree with D. tolerate E. settle with
3. The eldest son took a foolish decision which made him ... his claim to the traditional title A. neglect B. yield C. disclaim D. forfeit E. neglected

4. The president's speech at 7pm yesterday. A. is broadcast B. has been broadcast C. were broadcast D. was broadcast E. have been broadcast
5. The students were advised to look difficult words in the dictionary. A. in B. on C. out D. into E. Up
6. The building because of weak structural foundation. A. tumbled B. succumbed C. somersaulted D. collapsed E. collapsed

7. Since the writer did not indicate his name, the editor decided not to publish such.....article A. a discourteous B. an anonymous C. a cowardly D. a libellous E. unfriendly
8. He was charged with complicity.....the abortive coup. A. in B. for C. about D. on E. with
9. Of what he said made no sense. A. much B. majority C. plenty D. many E. much
10. things she had in the room were thrown out. A. so few B. the few C. all few D. very few E. a few
11. Pure solvents are obtained by A. distillation B. condensation C. extraction D. evaporation E. dissolution
12. Environmental pollution is worsened by the release from the automobile exhaust of A. water vapour B. steam C. smoke D. heavy metals E. steam and smoke
13. Sieving is a technique used to separate mixtures containing solid particles of A. small sizes B. large sizes C. different sizes D. the same size E. one size
14. Nitrogen obtained from the liquefaction of air has a higher density than that obtained from nitrogen containing compounds because the former contains A. water vapour B. oxygen C. carbon (iv) oxide D. rare gases
15. Hydrogen is used in oxy-hydrogen flames for melting metals because it A. involves a lot of heat when burnt B. combines explosively with oxygen C. is a very light gas D. is a rocket fuel E. it combines with air
16. Polyvinyl chloride is used to produce A. bread B. pencils C. ink D. pipes E. pots
17. The addition of water to calcium oxide leads to A. a physical change B. a chemical change C. the formation of a mixture D. an endothermic change
18. The metal that will react with water only in the form of steam to liberate hydrogen gas is A. calcium B. aluminum C. iron D. zinc E. platinum
19. 35 cm³ of hydrogen was sparked with 12 cm³ of oxygen at 110°C and 760 mmHg to produce steam. What percentage of the total volume of gas left after the reaction is hydrogen? A. 11% B. 3.31% C. 35% D. 69% E. 80%
20. In the extraction of iron in the blast furnace, lime stone is used to A. release CO₂ for the reaction B. reduce the iron ore C. increase the strength of the iron D. remove impurities
21. The force with which an object is attracted to the earth is its A. acceleration B. mass C. gravity D. weight E. momentum
22. A train has an initial velocity of 44 m/s and an acceleration of -4 m/s². Its velocity after 10 s is A. 2 m/s B. 4 m/s C. 8 m/s D. 12 m/s E. 16 m/s
23. A man of mass 50 kg ascends a flight of stairs 5 m high in 5 seconds. If the acceleration due to gravity is 10 m/s², the power expended is A. 500 W B. 400 W C. 250 W D. 200 W E. 100 W
24. Longitudinal waves do not exhibit A. refraction B. reflection C. diffraction D. polarization E. disaggregation
25. The unit of quantity of electricity is called A. amperes B. the volt C. the coulomb D. the ammeter
26. Which of the following is NOT a fundamental SI unit? A. metre B. ampere C. Kelvin D. radian
27. A simple pendulum with a period of 2.0 s has its length doubled. Its new period is A. 1.00 s B. 1.41 s C. 0.35 s D. 2.03 s E. 2.83 s
28. Which of the following will convert a millimeter to a voltmeter? A. low series resistance B. low parallel resistance C. high series resistance D. high parallel resistance E. medium parallel resistance
29. The distance between the fixed points of a centigrade thermometer is 20 cm. What is the temperature when the mercury level is 4.5 cm above the lower mark? A. 22.5°C B. 29.0°C C. 90.0°C D. 100.0°C E. 110.0°C
30. The amount of heat needed to raise the temperature of 10 kg of copper by 1 K is its A. specific heat capacity B. heat capacity C. latent heat D. internal heat E. specific latent heat
31. If $\sin(x+30)^\circ = \cos 40^\circ$ find x. A. 10° B. 3.20° C. 59° D. 65° E. 80°
32. Find the 9th term of the arithmetic progression, 18, 12, 6, 0, 6 A. -54 B. -30 C. 30 D. 42 E. 54
33. Solve the equation $4/a + 1/5a = 3/2$ A. 12/5 B. 11/3 C. 14/15 D. 24/5 E. 24/5
34. What is the number whose logarithm to base 10 is 3.47712? A. 3.0 B. 0.3 C. 0.03 D. 0.0003 E. 0.00003
35. The nth term of a sequence is given by $(-1)^{n-1}P^2z - 1$, find the sum of the second and third terms. A. 2B. 1C. 2D. 6E. -1
36. Evaluate $\log_{10} 25 + \log_{10} 32 - \log_{10} 8$. A. 0.2 B. 2C. 100 D. 409 E. 4
37. Which is the value of x satisfying the equation $4^{2x}/4^2 = 2$? A. -2 B. KC. K0.2 E. 3
38. Solve for x if $25^x + 3(5^x) = 4A. 1 \text{ or } -4 B. 0 C. 1 D. -4 \text{ or } 0 E. -1$
39. Teams P and Q are involved in a game of football. What is the probability that the game ends in a draw. A. % B. 1/3 C. % D. 2/3 E. 2/5
40. Find the value of m if $13m + 24m = 41m$. A. 8 B. 6 C. 50.2 E. 6
41. Hydra removes undigested food by A. passing it through the anus B. passing it through the mouth C. means of a contractile vacuole D. egesting it through the body surface E. passing it through the nose
42. A feature which adapts birds to flight is the possession of A. scaly legs B. light bones C. two walking legs D. a pointed beak E. combs
43. Physiological adaptation to very dry conditions in animals is called A. hibernation B. aestivation C. rejuvenation D. xeromorphism
44. The organelle involved in tissue respiration is the A. endoplasmic B. ribosome C. golgi body D. mitochondrion
45. The blood vessel which carries blood from the alimentary canal to the liver is A. hepatic artery B. hepatic vein C. hepatic portal vein D. mesenteric E. hepatic mesenterism
46. Soil with the finest particles is called A. silt B. clay C. sand D. gravel E. latrite
47. Interneural chlorosis is normally associated with the deficiency of A. magnesium B. potassium C. iron D. calcium E. iodine
48. In Amoeba, osmoregulation is carried out by the A. pseudopodium B. food vacuole C. contractile vacuole D. nucleus
49. In animals, meiosis comes A. after fertilization B. after every mitotic division C. before fertilization D. before every mitotic division E. during fertilization
50. Which of the following is transmitted through mosquito bite A. filariasis B. typhus C. plague D. schistosomiasis E. leprosy

LUTH SCHOOL OF NURSING PAST QUESTIONS AND ANSWERS

TEST 2019

TIME:35 MIN

Answer all questions:shade the answer sheet as appropriate with HG pencil only.

CHEMISTRY

- 1.A mixture contains 20cm³of hydrogen,35cm³of oxygen, 15cm³of carbon dioxide and 10cm³of nitrogen at S.T.P, which of the following gives the mole fraction of hydrogen in this mixture?A.0.02 B.0.16 C.0.20 D.0.25 E.20
2. 0.07g of a hydride of carbon occupies 56 at S.T.P when vapourised and contains 14.29%by mass hydrogen.The formula of the hydrocarbon is A.CH₄B.C₂H₂C.C₂H₄D. C₂H₆E.C₃H₈[C=12,H=1]
- 3.The pressure on 100cm³of oxygen gas at 35°C is 750mm of Hg.What would be the volume of the gas if the pressure is increased to 1000mm of Hg without changing the temperature.A.133.3cm³B.85cm³C.75cm³D.65cm³ E.58cm³
4. Which of the following bonds exist in crystalline ammonium chloride(NH₄Cl)?A ionic and covalent B.ionic and co-ordinate C.ionic,covalent and co-ordinate D.covalent,co-ordinate and metallic E.ionic,covalent and metallic
- 5.Which of the following is a neutralization reaction?
Addition of A.nitric acid to hydrochloric acid S.nitric acid to sulphuric acid C.acid to distilled water D.nitric acid to sodium hydroxide E.sodium chloride to distilled water
6. In the preparation of carbon monoxide by heating ethanedioic acid with concentrated sulphuric acid,the concentrated sulphuric acid acts as A.oxidizing agent B. reducing agent C.dehydrating agent D.reaction medium E.catalyst
- 7.How many grammes of methyl acetylene (propane) CH₃-C≡CH will completely discharge the colour of 8 g of bromine?(Br=80C=12,H=1)A.0.5B.1.0C.2.0 D.3.0 E.4.0
8. Brass is an alloy containing copper and A zinc B.tin C.aluminum D.silver E.lead
9. 60cm³of hydrogen are sparked with 20cm³of oxygen at 100°C and 1 atmosphere.The total volume of the residual gases is A.60cm³B.10cm³C.40cm³D.30cm³E.70cm³
- 10.If the rate of diffusion of oxygen gas is taken as 1,what will be the rate of diffusion of methane whose relative molar mass is 16?A.2.0 B.1.8C.1.4 D.1.0 E.0.5

USE OF ENGLISH

In each of questions 11 to 20,choose the word(s)or phrase(s) which best fill(s)the gap(s)

- 11.The sea wave continue tothe cliff on the west coast constantly A.impair B.rub C.knock D.erode
- 12.The college bus was traveling at a highwhen the accident occurred A.velocity B.acceleration C.rapidity

D.speed

- 13.Note that only senior members of staff have the of using the toilet upstairs A.permission B.occasion C. privilege D.habit
- 14.The chief priest will.....the main into the cut today. A.indoctrinate B.usher C.convert D.initiate
- 15.Obi is noted for hisattitude to his seniors at school.A. receptive B.respectful C.respective D.respectable
- 16.The girl that my brother introduced to us last week is prettyi-mannered A.and B.but also C.as well as D. respectable
- 17.The police report was.....to that of the eye witness.A. contrary B.inconsistent C.different D.congruent
- 18.The African extended family system gives security to..... members.A.his B.her C.its D.their
- 19.I know|.....read more but I am tired A.may B.ought to C.would D.could
- 20.Insects can becometo insecticides A.immured B.resistant C.Retient D.immobilized

PHYSICS

- 21.Which of the following is a scalar quantity?A.momentum B.acceleration C.displacement D.distance E.force
- 22.What change in velocity would produce a body of mass 4kg if a constant force of 16N acts on it for 2s?A.0.5m/s B.2.0m/sC.8.0m/sD.32.0m/sE.128.0m/s
- 23.A body accelerates uniformly from rest at the rate of 3m/s² for 8s.calculate the distance covered by the body during the acceleration.A.12m B.24mC.48m D.72m E.96m
- 24.Which of the following has the same unit as the moment of a force?A.force B.power C.Work D.momentum E.charge
- 25.Which of the following will reduce the frequency of oscillation of a simple pendulum?A.increasing the mass of the bob B.decreasing the mass of the bob C.increasing the length of the string D.decreasing the length of the string E.increasing the amplitude of oscillation
- 26.A barometer can be used in determining the length of a i. mountain ii.Depth of a mine iii.Dew point.Which of the following is/are correct?A.i,H,i B.H and ii only C.i and i only D.i and ii only E.iii only
- 27.Which of the following colours of surfaces will radiate heat energy best?A.red B.white C.black D.yellow E.blue
- 28.A gas which obeys Charles law exactly has a volume of 283cm³at 10°C,what is its volume at 30°C?A.142cm³ B.293cm³C.303cm³D.566cm³E.849cm³

29. A real image of an object formed by a converging lens of focal length 15cm is 3 times the size of the object. What is the distance of the object from the lens? A. 30cm B. 25cm C. 20cm D. 15cm E. 10cm
30. How far from a cliff should a boy stand to hear the echo of his clap 0.9s later? (speed of sound in air = 330m/s) A. 36.67m B. 74.25m C. 148.5m D. 297.00m E. 366.67m

MATHEMATICS

31. Find $\log_{10} 34$ if $\log_{10} 2 = 0.3010$ A. 5 B. 6 C. 7 D. 8
32. The radius of a circle is given as 5cm subject to an error of 0.1cm. What is the percentage error in the area of the circle? A. 1/25 B. % C. 4 D. 25
33. What is the value of x satisfying the equation $4^{2/4} = 2$ A. -2 B. -MC AD. 2
34. If $f(x) = 3 - \sqrt{x}$, find $x^2 + 36/x^2$ A. 9 B. 18 C. 24 D. 27
35. Solve the equation $y - 11y + 24 = 0$ A. 8, 3 B. 6, 4, 9 C. 6, 4 D. 9, -8
36. Aman invested a sum of N280.00 partly at 5% and partly at 4%. If the total interest is N12.80 per annum, find the amount invested at 5%. A. N14.00 B. N120.00 C. N140.00 D. N160.00
37. Ice forms on a refrigerator ice box at the rate of 4.06g per minute after 1 minute. If initially there were 2g of ice, find the mass of ice formed in 5 minutes A. 19.5 B. 17.0 C. 14.5 D. 12.5
38. Obtain a maximum value of the function $f(x) = x^3 - 12x + 11$ A. -5 B. -2 C. 2 D. 27
39. Two perfect dice were thrown together. Determine the probability of obtaining a total score of 8. A. 1/12 B. 5/36 C. 1/6 D. 7/36
40. The probability of an event P is $\frac{1}{6}$ while that of another event Q is $\frac{1}{12}$. If the probability of both P and Q is $\frac{1}{12}$, what is the probability of either P or Q? A. $\frac{1}{96}$ B. $\frac{1}{8}$ C. $\frac{5}{6}$ D. $\frac{11}{12}$

BIOLOGY

41. Which of the following organelles is used for locomotion in paramecium? A. pseudopodium B. irichocyst C. cilium D. pellicle E. contractile vacuole
42. Which of the following is not true of the nucleus of a living cell? It contains A. Chromosomes B. nucleolus C. nucleoplasm D. chromatids E. ribosomes
43. The cell membrane consists of A. carbohydrates and lipids B. vitamins and proteins C. lipids and proteins D. water and sugar E. starch and cellulose
44. Which of the following is not likely to be found in the cell of a ripe tomato fruit? A. plastids B. chlorophyll C. cellulose cell wall D. mitochondrion E. mineral salts
45. Osmosis can be defined as diffusion of A. atoms and molecules through a membrane to an area of higher concentration B. water molecules from a dilute solution to a concentrated solution across a permeable membrane C. water molecules from area of high concentration to an area of low concentration D. water molecules from a dilute solution to a concentrated solution through a semi-permeable membrane E. perspiration and excretion
46. The movement of diaphragm is a characteristic of gaseous exchange in A. insect B. fish C. toad D. mammal E. plants
47. In cellular respiration, energy is stored in the form of A. adenosine di-phosphate (ADP) B. adenosine mono-phosphate (AMP) C. adenosine tri-phosphate (ATP) D. heat energy E. electrical energy
48. The medium in which dissolved nutrients are transported in the body of vertebrates is called A. latex B. urine C. cell sap D. blood E. haemoglobin
49. Which of the following structures of the leaf contains air? A. guard cell B. palisade layer C. intercellular space D. vascular bundle E. upper epidermis
50. Which of the following organs is specially adapted for gaseous exchange in aquatic organisms? A. lungs B. trachea C. gills D. tracheoles E. Alveoli

SOLUTION TO LUTH SCHOOL OF NURSING 2019 EXAM

1. D	2. *	3. C	4. C	5. 0	6. C	7. C	8. A	9. E	10. D
11. D	12. D	13. A	14. D	15. D	16. B	17. A	18. C	19. B	208
21. C	22. C	23. E	24. C	25. C	26. E	27. C	28. C	29. A	30. C
31. A	32. A	33. B	34. C	35. A	36. D	37. D	38. D	39. B	40. C
41. C	42. D	43. C	44. B	45. D	46. D	47. C	48. D	49. C	50. C

LUTH SCHOOL OF NURSING PAST QUESTIONS AND ANSWERS

TEST 2018

THURSDAY, 29TH JULY 2017 TIME: 35 MIN

- Shadows and eclipses result from the [A] refraction of light [B] rectilinear propagation of light [C] diffraction of light [D] reflection of light.
- An object, which is 3cm high is placed vertically 10cm in front of a concave mirror. If this object produces an image 40cm from the mirror, the height of the image is [A] 0.75cm [B] 4.00cm [C] 8.00cm [D] 12.00cm
- A boy looks at the image of an object in a plane mirror. He observes two images, a main bright one and the other faint. The observed images result from [A] reflection only [B] refraction only [C] diffraction and interference [D] reflection and refraction.
- What must be the distance between an object and a converging lens of focal length 20cm to produce an erect image two times the object height? [A] 20cm [B] 15cm [C] 10cm [D] 5cm.
- For a shortsighted person, light rays from a point on a very distant object are focused [A] in front of the retina [B] on the retina by a converging lens [C] behind the retina by a diverging lens [D] in front of the retina at a distance of 2F from the lens
- When light is incident on an object, which is magenta in

- color, which of the following colors would be absorbed? [A] red and blue [B] green only [C] red and green [D] red only.
7. In a resonance tube experiment, the effective length of the air column for the first resonance is 20 cm when set into vibration by a tuning fork of frequency 480 Hz. Neglecting end effect, the velocity of sound in air is [A] 96 m/s [B] 255 m/s [C] 340 m/s [D] 384 m/s,
8. A sonometer wire of length 100 cm under a tension of 10 N, has a frequency of 250 Hz. Keeping the length of the wire constant, the tension is adjusted to produce a new frequency of 350 Hz. The new tension is [A] 5.1 N [B] 7.1 N [C] 14.0 N [E] 19.6 N.
9. One of the properties of the earth's magnetic field is that the [A] north pole lies in the northern hemisphere [B] geographic and magnetic meridians coincide [C] earth's magnetic flux is entirely horizontal at a place where the magnetic dip is zero [D] earth's magnetic flux is entirely vertical at a place where the magnetic dip is zero.
10. Three cells each of e.m.f. 1.5 V and an internal resistance of 1.0 are connected in parallel across a load of resistance 2.67. Calculate the current in the load [A] 0.26 A [B] 0.41 A [C] 0.50 A [D] 0.79 A.
11. A government spokesman announced that efforts..... the release of the hostages are continuing [A] to obtain [B] in obtaining [C] for obtaining [D] of obtaining.
12. I know you think I am talking nonsense, Shehu but..... you'll realize that I was right [A] at one time [B] on time [C] in time [D] at times.
13. The inspector of Education who made several trips on the bad road returned yesterday completely..... by fever [A] brought down [B] put down [C] worn down [D] worn off.
14. The vice principal asked the students to always their answers only from the textbooks recommended for the course [A] look out [B] search out [C] look up [D] bring up.
15. I know that your friend will not accept the proposal..... [A] and you neither [B] and neither you [C] neither do you [D] neither will you.
16. Mark is a very handsome fellow who informs me that he has for pretty girls [A] a heart [B] a lip [C] an eye [D] a check.
17. Wale Agun, in creating his characters, draws freely..... his experience in life [A] by [B] in [C] on [D] of.
18. When I have an appointment with someone, I hate waiting [A] to be keeping [B] for being kept [C] being kept [D] in being kept.
19. It's no good about the result until you have sat for the examination [A] to worry [B] for worrying [C] worrying [D] to have worried.
20. If you don't want to your car to robbers, then don't travel in the night [A] loose [B] loss [C] lose [D] lost.
21. The number 25 when converted from the tens and units base to the binary base (base two) is one of the following.
[A] 10011 [B] 111011 [C] 111000 [D] 11001 [E] 110011.
22. Evaluate $(6.3 \times 10^3) / (81 \times 10^3)$ to 3 significant figures [A] 77.80 [B] 778.0 [C] 7.870 [D] 8.770 [E] 188.70.
23. The positive root of t in the following equation $4t^2 + 7t - 1 = 0$, correct to 4 places of decimal, is [A] 1.0622 [B] 10.6225 [C] 0.1328 [D] 0.3218 [E] 2.0132.
24. The difference between the length and width of a rectangle is 6 cm and the area is 135 cm². What is the length? [A] 25 cm [B] 18 cm [C] 15 cm [D] 24 cm [E] 27 cm.
25. The first term of an Arithmetic Progression is 3 and the fifth term is 9. Find the number of the terms in the progression if the sum is 81. [A] 12 [B] 27 [C] 9 [D] 4 [E] 36.
26. The difference between 4^7 and 2^4 greater than the sum
14
[E] 48/56.
27. Multiply $2^2 + 1$ by $2^2 + 1$. [A] $4^2 + 32$ [B] $t^2 - 1$ [C] $4^2 - 6$ [D] $4^2 - 4 + 1$ [E] $4^2 - 32 + 1$
28. A baking recipe calls for 2.5 kg of sugar and 4.5 kg flour with this recipe some cakes were baked using 24.5 kg of a mixture of sugar and flour how much sugar was used? [A] 12.25 g [B] 6.5 kg [C] 8.75 kg [D] 15.75 kg [E] 8.25 kg.
29. The sum of the root of a quadratic is 5/2 and then product of its root is 4, the quadratic equation is [A] $2^2 + 5 + 8 = 0$ [B] $2^2 - 5 + B = 0$ [C] $2^2 - 8 + 5 = 0$ [D] $2^2 + 8 - 5 = 0$ [E] $2^2 - s + B = 0$.
30. Solve the given equation $(\log a)^2 - 6 \log s + 9 = 0$ [A] 27 [B] 9 [C] 3/z [D] 18 [E] 81.
31. In which order are the following salts sensitive to light? [A] AgI AgCl AgBr [B] AgCl AgI AgBr [C] AgBr AgCl AgI [D] AgCl AgBr AgI.
32. The pOH of a solution of 0.25 mol dm³ of hydrochloric acid is [A] 12.40 [B] 13.40 [C] 14.40 [D] 14.60.
33. $\text{MnO}_2 + 8\text{H}^+ + 2\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$ in the equation represents [A] 2e [B] 3e [C] 5e [D] 7e
34. Given that M is the mass of substance deposited in an electrolysis and Q the quantity consumed, then Faraday's law can be written as [A] Q/Z [B] Q/Z [C] Z/Q [D] M/Q .
35. 0.46 g of ethanol when burned raised the temperature of 50 g of water by 14.3 K. Calculate the heat of combustion of ethanol. [A] +33000 KJ mol⁻¹ [B] +300 KJ mol⁻¹ [C] 300 KJ mol⁻¹ [D] 300 KJ mol⁻¹ $c = 12.0$ $H = 1$, Specific heat capacity of water = 4.2 JgK
36. Powdered marble reacts faster with hydrochloric acid solution than the granular form because the powdered form has [A] more molecules [B] more atoms [C] larger surface area [D] relatively large mass.
37. For a reaction in equilibrium, the species involved in the equilibrium constant expression are [A] gaseous and solid species [B] liquid and solid species [C] solid and dissolved species [D] gaseous and dissolved species.
38. A phenomenon where an element exists in different forms in the same physical state is known as [A] isomerism [B] amorphism [C] allotropy [D] isotropy.
39. The substance often used for vulcanization of rubber is [A] chlorine [B] hydrogen peroxide [C] sulphur [D] tetraoxosulphate (VI) acid.
40. A gas that is not associated with global warming is [A] CO₂ [B] SO₂ [C] CH₄ [D] H₂.
41. Which of the following structures is capable of producing more tissues in the stem of a herbaceous flowering plant? [A] Epidermis [B] pericycle [C] xylem [D] cambium.
42. The manufacture of carbohydrates by plants takes place only in [A] the leaves [B] the green stems [C] chlorophyllous parts [D] flowering plants.
43. In water culture experiment, a plant showed poor growth

and yellowing of the leaves. These may be due to deficiency of [A]copper [B]iron [C]magnesium [D]calcium.

44. In Millon's test, when the reagent is added to a protein food item, a white precipitate is produced which turns [A]blue on heating [B]yellow on heating [C]green on heating [D]red on heating.

45. Regulation of blood sugar level takes place in the [A]pancreas [B]ileum [C]liver [D]kidney.

46. Unicellular organisms transport essential nutrients directly to all parts of their bodies by the process of diffusion because they have [A]a large volume to surface area ratio [B]a large surface area to volume ratio [C]their bodies immersed in the nutrients [D]their outer membrane made of cellulose.

47. The heart of the adult frog consists of [A]two auricles and two ventricles [B]one auricle and one ventricle [C]two ventricles and one auricle [D]one ventricle and two auricles.

48. In adult mammalian blood, the cells, which lack nuclei, are the [A]erythrocytes [B]lymphocytes [C]leucocytes [D]phagocytes.

49. Which of the following movements occur during exhalation? [A]the diaphragm and intercostal muscles relax [B]the thoracic cavity increases in volume [C]the diaphragm and intercostal muscles contract [D]the diaphragm contracts and the intercostal muscles relax.

50. In which of the following groups of animals is the Malpighian tubule found? [A]lizards, snakes and frogs [B]crickets, houseflies and grasshoppers [C]millipedes, centipedes and scorpions [D]earthworms, roundworms, and flatworms.

SOLUTION TO LUTH SCHOOL OF NURSING 2018 EXAM

1. B	2. D	3. A	4. *	5. A	6. B	7. D	8. E	9. C	10. B
11. A	12. C	13. C	14. C	15. D	16. B	17. A	18. C	19. C	20. A
21. D	22. *	23. C	24. C	25. C	26. E	27. B	28. C	29. B	30. A
31. D	32. C	33. C	34. D	35. A	36. C	37. D	38. D	39. C	40. D
41.								49. A	50. B

LUTH SCHOOL OF NURSING PAST QUESTIONS AND ANSWERS

TEST 2017

Sat. 4th July 2016 TIME: 1 Hour

- Which of the following is not an example of a force? (A)tension (B)Weight (C)friction (D)mass (E)thrust
- A body moves along a circular path with uniform angular speed of 0.6 rad s^{-1} and at a constant speed of 3.0 m s^{-1} calculate the acceleration of the body towards the centre of the circle (A) 25.0 m/s^2 (B) 5.4 m/s^2 (C) 5.0 m/s^2 (D) 1.8 m/s^2 (E) 0.2 m/s^2
- Which of the following is a derived unit? (A)Ampere (B)Kilogram (C)second (D)Ohm (E)Kelvin
- A boy timed 20 oscillations of certain pendulum three times and obtained 44.3s, 45.5s and 45.7s respectively. Calculate the mean period of oscillations of the pendulum (A)0.13s (B)2.22s (C)2.26s (D)44.30s (E)45.17s
- A particle starts from rest and moves with a constant acceleration of 0.5 m/s^2 . Calculate the time by the particle to cover a distance of 25m (A)2.5s (B)7.1s (C)10.0s (D)50.0s (E)100.0s
- A block of material of volume $2 \times 10^{-5} \text{ m}^3$ and density $2.5 \times 10^3 \text{ kg/m}^3$ is suspended from a spring balance with half the volume of the block immersed in water. What is the reading of the spring balance? (Density of water = $1.0 \times 10^3 \text{ kg/m}^3$, $g = 10 \text{ m/s}^2$) (A)0.10N (B)0.10N (C)0.30N (D)0.40N (E)0.50N
- An object is projected with a velocity of 100 m/s from the ground level at an angle θ to the vertical. If the total time of flight of the projectile is 10s, calculate θ ($g = 10 \text{ m/s}^2$) (A)0 (B)30 (C)45 (D)60 (E)90
- How far will a body move in 4 seconds if uniform acceleration from rest at rate of 2 m/s^2 ? (A)32m (B)24m (C)16m (D)12m (E)8m
- If the temperature of water is gradually increased from 0 to 40°C , the density of the water within this range (A)increase

for a while and then decreases (B)decreases for a while and then increases (C)increases gradually (D)decreases gradually (E)remains the same

- The expansion of solids can be considered a disadvantage in the (A)fire alarm system (B)thermostat (C)riveting of steel plates (D)balance wheel of a watch (E)fitting of wheels on rims
- A solid metal cube of side 10cm is heated from 10°C to 60°C . If the linear expansivity of the metal is $1.2 \times 10^{-5} \text{ K}^{-1}$, calculate the increase in its volume (A) 0.0 cm^3 (B) 1.2 cm^3 (C) 18 cm^3 (D) 3.6 cm^3 (E) 6.0 cm^3
- A gas has a volume of 546 cm^3 at 0°C . What is the volume of the gas at 100°C if its pressure remains constant? (A) 346 cm^3 (B) 446 cm^3 (C) 546 cm^3 (D) 646 cm^3 (E) $\infty \text{ cm}^3$
- An image which cannot be formed on a screen is said to be (A)inverted (B)real (C)erect (D)blurred
- Longitudinal waves cannot be (A)diffracted (B)refracted (C)polarized (D)reflected (E)superposed
- The images formed by a diverging lens are always (A)diminish, virtual and inverted (B)diminished, virtual and erect (D)magnified, virtual and erect (E)magnified, real and inverted
- In the normal use of a simple microscope, a person sees an (A)inverted, virtual and magnified image (B)erect, virtual and magnified image (C)erect, real and magnified image (D)magnified, real and magnified image (E)inverted and real image the same as the object
- A lens of focal length 15.0 cm forms an upright image four times the size of an object. Calculate the distance of the image from the lens (A) 11.3 cm (B) 18.8 cm (C) 37.5 cm (D) 45.0 cm (E) 75.0 cm

18. An object is placed between two mirrors which are inclined at an angle of 120° and facing each other. Determine the number of images observed in two mirrors. (A) 1 (B) 2 (C) 3 (D) 4 (E) 5
19. In a ripple tank experiment, a vibrating plate is used to generate ripples in the water. If the distance between two successive troughs is 3.5 cm and the wave travels a distance of 31.5 cm in 1.5 s, calculate the frequency of the vibrator. (A) 30 Hz (B) 6.0 Hz (C) 12.0 Hz (D) 27.0 Hz (E) 73.5 Hz
20. Which of the following have the longest wave lengths? (A) infra-red rays (B) gamma rays (C) x rays (D) ultraviolet rays (E) radio waves
21. Simplify $12.5^{1/3} \times 49^2$. (A) 350 (B) 35 (C) $1/35$ (D) $\sqrt{350}$ (E) 0
22. If $3=2$, what is x ? (A) 1 (B) 1.5 (C) 4.5 (D) 18 (E) 40.5
23. Express 0.00562 in standard form. (A) 5.62×10^3 (B) 5.62×10^2 (C) 5.62×10^1 (D) 5.62×10^0 (E) 5.62×10^{-3}
24. Given that $\log_{10} P = 1$, find the value of P . (A) $1/10$ (B) 0.3 (C) 10 (D) 100 (E) 1000
25. Simplify: $\log_a(\log_{10}(A^{2/3}(B)^{\%}) / \log_2(D)^3) \log_2$
26. If $\log x = 2.3675$ and $\log y = 0.9750$, find $x+y$, correct to three significant figures? (A) 1.18 (B) 9.31 (C) 9.03 (D) 9.44 (E) 9.46
27. While doing his physics practical, Idowu recorded a reading as 1.12 cm instead of 1.21 cm. Calculate his percentage error. (A) 1.17% (B) 6.38% (C) 7.44% (D) 8.05% (E) 9.00%
28. Find the 4th term of an A.P. whose first term is 2 and the common difference is 0.5. (A) 0.5 (B) 2.5 (C) 3.5 (D) 4 (E) 4.5
29. An arc of length 22 cm subtends an angle of θ at the centre of the circle. What is the value of θ if the radius of the circle is 15 cm? (Taken $\pi = 22/7$) (A) 70 (B) 84 (C) 96 (D) 156 (E) 168
30. Find the sum of the first five terms of the GP 2, 6, 18, ... (A) 484 (B) 243 (C) 242 (D) 130 (E) 121
31. Let J be the set of positive integers. If $H = \{x \in J : x^2 < 3 \text{ and } x \neq 0\}$ then (A) $H = \{1\}$ (B) H is an infinite set (C) $\{0, 1, 2\} \subset H$ (D) $H = \emptyset$ (E) $J \subset H$
32. In a class of 80 students, 40 had to study Economics or Geography, or both Economics and Geography. If 64 students studied Economics and 50 studied Geography, how many studied both subjects? (A) 15 (B) 30 (C) 35 (D) 45 (E) 50
33. Factorize $x^2 + 4x - 192$. (A) $(x-4)(x+48)$ (B) $(x+1)(x-12)$ (C) $(x-12)(x+16)$ (D) $(x-12)(x+16)$
34. Factorize $2e^2 - 3e + 1$. (A) $(2e-1)(e-1)$ (B) $(e+1)(2e+1)$ (C) $(2e+3)(3-2e)$ (D) $(2e-3)(e-1)$ (E) $(e-2)(3-2e)$
35. Solve the equation $7y^2 = 3y$. (A) $y = 3$ or 7 (B) $y = 0$ or 7 (C) $y = 0$ or 9 (D) $y = 0$ or 10
36. Solve the equation $2a^2 - 3a - 27 = 0$. (A) $3/2, 9$ (B) $-2/3, 9$ (C) $3, 9/2$ (D) $-3, -9/2$ (E) $-3, 9/2$
37. A sector of a circle of radius 7 cm has an area of 44 cm^2 . Calculate the angle of the sector, correct to the nearest degree. (A) 6 (B) 26 (C) 52 (D) 103 (E) 206
38. If the shadow of a pole 7 m high is $1/2$ its length, what is the angle of elevation of the sun, correct to the nearest degree? (A) 90 (B) 63 (C) 60 (D) 26 (E) 0
39. From the top of a building 10 m high, the angle of depression of a stone lying on the horizontal ground is 69° . Calculate, correct to 1 decimal point, the distance of the stone from the foot of the building. (A) 3.6 m (B) 3.8 m (C) 6.0 m (D) 9.3 m (E) 26.1 m
40. The bearing of a point X from Y is 074° . What is the bearing of Y from X? (A) 106° (B) 148° (C) 164° (D) 254° (E) 286°
41. The nucleus is considered the central organelle of a cell because it (A) contains the genetic material (B) contains the nuclear sap (C) is bounded by the nuclear membrane (D) is located at the centre of the cell
42. In a prokaryotic cell, the type is characterized by (A) complex cytoplasm in which different regions are poorly defined (B) localization of different regions of the cell into tissues (C) collection of organelles and macromolecular complexes (D) simple cytoplasm with well-defined regions
43. The natural tendency of an organism as they evolve is to (A) decrease in size (B) increase in size (C) develop specialized structures (D) feed indiscriminately
44. In snails, the hard calcareous shells are secreted by the (A) radius (B) tentidium (C) pneumostome (D) mantle
45. The ability of the cockroach to live in cracks and crevices is enhanced by the possession of (A) wings and segmented body (B) compound eyes (C) claws on the legs (D) dorso-ventrally flattened body
46. The caste of termites that lacks pigmentation is the (A) king (B) worker (C) soldier (D) queen
47. The structures that prevent food particles from escaping through the fish gill are (A) arches (B) filaments (C) rakers (D) lamellae
48. A distinguishing feature of mammals is the possession of (A) skin (B) scale (C) nail (D) hair
49. Which of the following structures is capable of producing more tissues in the stem of a herbaceous flowering plant? (A) epidermis (B) pericycle (C) xylem (D) cambium
50. The manufacture of carbohydrates by plants takes place only in (A) the leaves (B) the green stems (C) chlorophyllous parts (D) flowering plants
51. In a water culture experiment, a plant showed poor growth and yellowing of the leaves. These may be due to deficiency of (A) copper (B) iron (C) magnesium (D) calcium
52. In Millon's test, when the reagent is added to a protein food item, a white precipitate is produced which turns (A) blue on heating (B) yellow on heating (C) green on heating (D) red on heating
53. Regulation of blood sugar level takes place in the (A) pancreas (B) ileum (C) liver (D) kidney
54. Unicellular organisms transport essential nutrients directly to all parts of their bodies by the process of diffusion because they have (A) a large volume to surface area ratio (B) a large surface area to volume ratio (C) their bodies immersed in the nutrients (D) their outer membrane made of cellulose
55. The heart of the adult frog consists of (A) two auricles and two ventricles (B) one auricle and one ventricle (C) two ventricles and one auricle (D) one ventricle and two auricles
56. In adult mammalian blood, the cells which lack nuclei are the (A) erythrocytes (B) lymphocytes (C) leucocytes (D) phagocytes
57. Which of the following movements occur during exhalation? (A) the diaphragm and intercostal muscles relax (B) the thoracic cavity increases in volume (C) the diaphragm and intercostal muscles contract (D) the diaphragm contracts and the intercostal muscles relax

58. In which of the following groups of animals is the Malpighian tubule found? (A) lizards, snakes and frogs (B) crickets, house flies and grasshoppers (C) millipedes, centipedes and scorpions (D) earthworms, roundworms and flatworms
59. Which of the following is not a function of the mammalian skeleton? (A) protection (B) respiration (C) transportation (D) support
60. The most reliable estimate of growth is by measuring changes in- (A) length (B) volume (C) surface area (D) dry weight
61. Which of the following is not a physical change? (A) the bubbling of chlorine into water (B) the bubbling of chlorine into a jar containing hydrogen (C) the dissolution of sodium chloride in water (D) the passing of steam over heated iron.
62. In the reaction $\text{SnO}_2 + 2\text{C} \rightarrow \text{Sn} + 2\text{CO}$, the mass of coke containing 80% carbon required to reduce 0.302 kg of pure tin oxide is (A) 40 kg (B) 0.20 kg (C) 0.06 kg (D) 0.04 kg.
(Sn=119, O=16, C=12)
63. The Avogadro number of 24 g of magnesium is the same as that of (A) 1 g of hydrogen molecules (B) 16 g oxygen molecules (C) 12 g of carbon molecules (D) 35.5 g of chlorine molecules.
64. A gas occupies a container of volume 146 cm³ at 180°C and 0.971 atm its volume in cm³ at STP (A) 133 (B) 146 (C) 266 (D) 292
65. The volume occupied by 1.58 g of a gas at STP is 500 cm³. What is the relative molecular mass of the gas? (A) 28 (B) 32 (C) 44 (D) 71 (G.M.V. at S.T.P. = 22.4 dm³)
66. Equal volumes of CO, SO₂, NO₂ and H₂S were released into a room at the same point and time, Which of the following gives the order of diffusion of the gases to the opposite corner of the room? (A) CO, SO₂, NO₂, H₂S (B) SO₂, NO₂, H₂S, CO (C) CO, H₂S, SO₂, NO₂ (D) CO, H₂S, NO₂, SO₂ (S=32, C=12, O=16, N=14, H=1)
67. A basic postulate of the kinetic theory of gases is the molecules of a gas move in straight lines between collisions. This implies that (A) collisions are perfectly elastic (B) forces of repulsion exist (C) forces of repulsion and attraction are in equilibrium (D) collisions are inelastic
68. Which of the following terms indicates the number of bonds that can be formed by an atom? (A) oxidation number (B) valence (C) atomic number (D) electronegativity
69. X(g) → Y(g) The type of energy involved in the above transformation is (A) ionization energy (B) sublimation energy (C) lattice energy (D) electron affinity
70. Chlorine, consisting of two isotopes of mass number 35 and 37, has an atomic mass of 35.5. The relative abundance of the isotopes of mass number 37 is (A) 20% (B) 25% (C) 50% (D) 75%
71. 10.0 dm³ of air containing H₂S as an impurity was passed through a solution of Pb(NO₃)₂ until all the H₂S had reacted. The precipitate of PbS was found to weigh 5.02 g. According to the equation: $\text{Pb}(\text{NO}_3)_2 + \text{H}_2\text{S} \rightarrow \text{PbS} + 2\text{HNO}_3$ the percentage by volume of hydrogen sulphide (A) 50.2 (B) 47.0 (C) 4.47 (Pb=207, S=32, G.M.V. at S.T.P. = 22.4 dm³)
72. A blue solid T, which weighed 5.0 g was placed on a table. After 8 hours the resulting pink solid was found to weigh 5.5 g it can be inferred that substance T (A) is deliquescent (B) is hygroscopic (C) has some molecules of water of

crystallization (D) is efflorescent

73. The effluent of an industrial plant used on electrolysis of concentrated brine, with a flowing mercury cathode may contain impurities like (A) oxygen (B) hydrogen (C) mercury (D) chloride (D) hydrogen chloride
74. The solubility in moles per dm³ of 20 g of CuSO₄ dissolved in 100 g of water at 180°C is (A) 0.13 (B) 0.25 (C) 1.25 (D) 2.00 (Cu=63.5, S=32, O=16)
75. Smoke consists of (A) solid particles dispersed in liquid (B) solid or liquid particles dispersed in gas (C) gas or liquid particles dispersed in liquid (D) liquid particles dispersed in liquid.
76. $\text{Na}_2\text{C}_2\text{O}_4 + \text{CaCO}_3 \rightarrow \text{CaC}_2\text{O}_4 + 2\text{NaCl}$. Given a solution of 1.9 g of sodium oxalate in 50 g of water at room temperature, calculate the minimum volume of 0.1 M calcium chloride required to produce maximum calcium oxalate using the above equation. (A) 1.40 × 10⁻² dm³ (B) 1.40 × 10⁻² dm³ (C) 1.40 × 10⁻² dm³ (D) 1.40 × 10⁻² cm³
77. 2.0 g of a monobasic acid was made up to 250 cm³ with distilled water. 25.00 cm³ of this solution required 20.00 cm³ of 1 M NaOH solution for complete neutralization of the molar mass of the acid is (A) 200 g (B) 150 g (C) 100 g (D) 50 g
78. What is the concentration of H⁺ ions in moles per dm³ of a solution of pH 4.398? (A) 4.0 × 10⁻² (B) 0.4 × 10⁻² (C) 4.0 × 10⁻³ (D) 0.4 × 10⁻³
79. What volume of 11.0 M hydrochloric acid must be distilled to obtain 1 dm³ of 0.05 M acid? (A) 4.54 cm³ (B) 5.65 cm³ (C) 6.76 cm³ (D) 7.87 cm³
80. If 10.8 g of silver is deposited in a silver coulometer connected in series with a copper coulometer, the volume of oxygen liberated is (A) 0.56 dm³ (B) 5.60 dm³ (C) 11.29 dm³ (D) 22.40 dm³ (Ag=108; Cu=64; G.M.V. at S.T.P. = 22.40 dm³)

Instruction: Choose the word opposite in meaning to the underlined words.

81. The young engineer is good at terminating other people's projects but has not been capable of ----any of his own.
(A) integrating (B) finishing (C) completing (D) initiating (E) organizing
82. The manager who is expected to be given respect was treated with----- (A) dignity (B) scorn (C) shame (D) cruelty (E) disloyalty
83. Those who had invitation cards were admitted to the party while those who had none were----- (A) barred (B) repelled (C) expelled (D) compelled (E) restricted
84. Nobody expects him to show -----for his children but he certainly bestows too much affection on them (A) love (B) concern (C) intimacy (D) devotion (E) hatred
85. The challenger was crude and inexperienced in contrast to the champion who was----- (A) great (B) exposed (C) celebrated (D) refined (E) strong

From the words lettered A to E, choose the word that best completes each of the following sentences.-

86. The fisherman threw a stone into the river and this caused a ----- (A) sprinkle (B) sparkle (C) splash (D) spring (E) storm
87. The play was so interesting that the -----clapped for quite a long time at the end. (A) spectators (B) watchers

-
- (C)congregation(D)people(E)audience
88.The building-----because of weak structural foundation.
(A)tumbled (B)succumbed (C)somersaulted (D)collapsed
(E)caved
89.The magezine was -----by the government for an
offensive publication.(A)prescribed(B)proscribed
(C)suspended (D)condemned (E)persecuted
90.Many people reacted to the brutal murder of the popular
journalist with strong -----(A)Indignation(B)demonstrate
(C)mobilization (D)candemnation (6)accusation

From the words or groups of words lettered A-E below each of the following sentences,choose the word or group of words that is nearest in meaning to the underlined word or group of words as used in the sentence.

- 91.It takes a great deal of stamina to run the marathon race.
(A)courage(B)determine (C)energy (D)intelligence (E)
cleveness
92.But for the principal actor the play would have been dull.
(A)Important (B)head (C)master (D)famous (E)main
93.An open carhas no protection against the
elements(A)weather (B)emergency (C)molecule

-
- 94.He was reluctant to grant my request (A)disposed
(B)delight (C)reticent {D)unwilling(E)agreeable
95.The detective was perplexed when the clues in the
murder case pointed to at least a dozen different suspects
(A)surprised(B)confused (C)excited (D)discouraged (E)
disappointed
96.The Military Governar called for a concerted effort in solving
the problems of the state.(A)a dramatic (B)an agitated (C)a
joint (D)a direct (E)an unfailing
97.My financial situation is so precarious that very scon I may
be insolvent(A)borrowing(B)steeling(C)soluble
(D)dependent (E)bankrupt
98.The chairman is of the opinion that accepting the
proposal would be inimical to the objectives of the
association.(A)harmful(B)relevant (C)irrelevant
(D)indispensable(E)helpful
99.The famous politician was noted for his pragmatic appraach
to national interest (A)idealistic (B)romantic (C)
compromising(0)practical (E)inconsistent
100.Kunle is very pessimistic about our chance of success (A)sad
(B)despondent (C)unconvinced (D)worried (E)concerned

(D)atoms(E)atmosphere

SOLUTION TO LUTH SCHOOL OF NURSING 2017 EXAM

1. D	2. D	3. B	4. C	5. C	6. B	7. B	8. E	9. C	10. C
11. E	12. A	13. C	14. C	15. C	16. B	17. D	18. D	19. *	20. C
21. C	22. B	23. A	24. E	25. B	26. B	27. E	28. C	29. ~	30. A
31. A	32. C	33. "	34. D	35. C	36. E	37. D	38. B	39. E	40. D
41. A	42. A	43. C	44. 4	45. D	46. B	47. C	48. D	49.	50. C
51. B	52. D	53. A	54. B	55. D	56. A	57. A	58. B	59. C	60. C
61. C	62.	63. C	64. A	65. D	66.	67. A	68. B	69.	70. B
71. C	72. B	73. C	74. C	75. B	76	77	78. A	79. *	80. °
81. D	82. B	83. A	84. E	85. D	86. C	87. E	88. 0	89. B	90. A
91. C	92. E	93. A	94. D	95. B	96. C	97. E	98. A	99. D	100. B